

# Mehrdad Momeni Zadeh

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## TECHNICAL SKILLS

**Languages:** Java, Python, C++, Go, TypeScript, SQL, R, Kotlin

**Web and Backend:** Spring Boot, Hibernate, Next.js, React, Django, RESTful APIs, GraphQL

**DevOps and Cloud:** AWS (S3, EC2), GCP, Docker, Kubernetes, Jenkins, Git, CI/CD, Kafka, JWT/OAuth2

**ML and Data:** PyTorch, TensorFlow, scikit-learn, OpenCV, WandB, Hydra, NumPy, Pandas

**Databases:** PostgreSQL, MySQL, MongoDB, Firebase, H2

**Testing Frameworks:** JUnit 5, Mockito, Testcontainers, pytest, testify, gomock

## TECHNICAL PROJECTS

**StagePass** — Full-stack event ticketing system for university events.

[GitHub](#) 🐙

*Tech:* Java, Spring Boot 3, Spring Security, TypeScript, PostgreSQL, React, Stripe, REST APIs

- Deployed for **SFU events (2025)**, processing **100+** student tickets with zero payment errors.
- Secured organizer and attendee flows via **Spring Security** and **OAuth2**, ensuring role-based access.
- Led a **team of 3** to design **REST APIs** and a **PostgreSQL** schema that reliably handled **300+** tickets per day.

**MicroGopher** — Distributed microservices platform in Go.

[GitHub](#) 🐙

*Tech:* Go, Docker, Kubernetes, PostgreSQL, MongoDB, gRPC, AMQP

- Split a monolith into modular **Go microservices**, improving fault isolation and scalability.
- Developed microservices for authentication, logging, and mailer communicating via **REST APIs** and **gRPC**.
- Deployed with **Docker** and **Kubernetes**, handling **10K+** requests per second with under **1%** errors.

**ShareScreenly** — Platform for screen recording and instant video sharing.

[GitHub](#) 🐙

*Tech:* TypeScript, Next.js 15, Node.js, PostgreSQL, Tailwind, AWS S3, FFmpeg

- Adopted by **100+** SFU students, cutting demo setup from **5 min** to **30 sec**.
- Integrated **AWS S3** and **FFmpeg** for storage, compression, and metadata; supported **1GB+** uploads.
- Shipped role-based access with **Better Auth** and a responsive **Tailwind** UI.

**HirePrep** — AI-powered mock interview platform with real-time voice practice.

[GitHub](#) 🐙

*Tech:* TypeScript, Next.js 15, Firebase, Gemini AI, Vapi, ShadCN, AWS S3

- Piloted with **50+** **SFU students**, improving interview scores by **20%** in surveys.
- Generated tailored interview questions and instant **AI-driven feedback** using Gemini.
- Deployed assets on **AWS S3** for scalable storage and delivery.

**NeuroDriver** — 2D driving simulator with neural agents.

[GitHub](#) 🐙

*Tech:* Python, Box2D, Genetic Algorithms, NumPy

- Trained neural agents to reduce collisions by **20%** over **50 generations**.
- Increased lap completion rates by **30%** through reward-tuned training.
- Automated training runs and created GIF replays to make experiments easy to track and repeat.

## WORK EXPERIENCE

**DeLTA Lab** — Simon Fraser University

Burnaby, BC

*Undergraduate Researcher* — Applied Machine Learning (🐙 [Code](#))

Jan 2025 – Aug 2025

- Built and deployed **ML training pipelines** with **Docker** to **Google Cloud TPU-v4**, cutting setup time by **85%**.
- Implemented automated **CI/CD workflows** in a **6-person** team, reducing integration bugs by **30%**.
- Optimized model runs to achieve **4× faster training** and **12% higher accuracy** on large-scale datasets.
- Collaborated across research and engineering teams to design scalable **data processing tools**.
- Co-authored a paper submitted to **NeurIPS 2025**, strengthening technical writing and teamwork skills.

## EDUCATION

**BSc. in Computer Science** — GPA: 3.4/4.33

Expected May 2027

*Simon Fraser University*

Burnaby, BC

- **Relevant Coursework:** Data Structures, Computer Systems, Databases, Networking, Machine Learning, Deep Learning